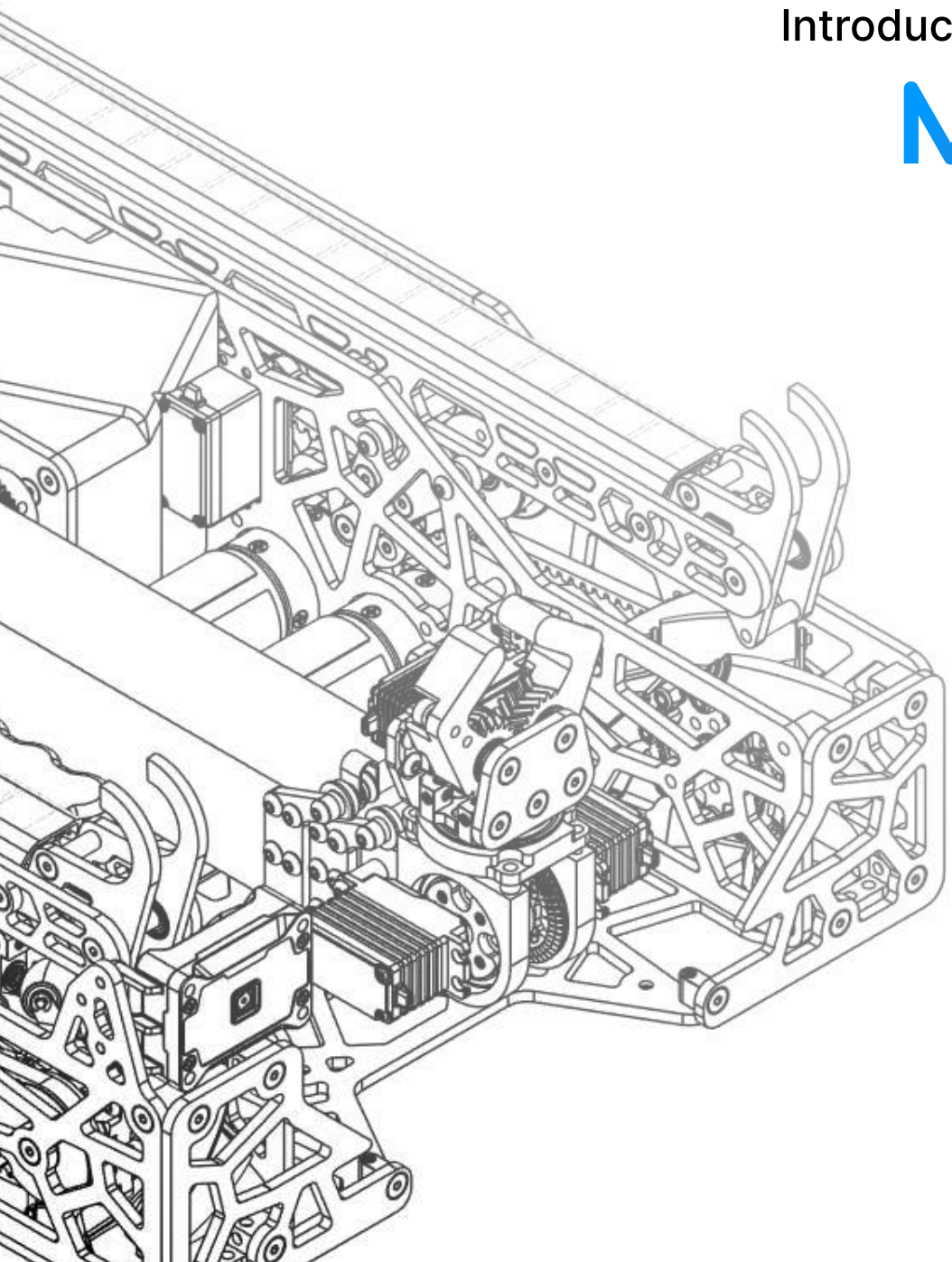
FTC #21579

TESTING IS OPTIONAL

Introducing

MAKO

Our 24-25 Season Robot



1. Introduction
2. Sustainability + Mentors
3. Community Outreach
4. Technical Outreach
5. FIRST Outreach
6. Game Strategy
7. Design Process
8. Overview of MAKO
9. Drivetrain
10. Arm
11. Manipulators
12. Ascent
13. Control Theory
14. TeleOp Automations
15. Autonomous

Introduction

About the Team

We are Testing is Optional, team 21579, community-based out of Tucson, Arizona. This season we have 3 adult mentors and 12 students, many are returning.

Our primary goal is to innovate and educate. We aim to innovate in hardware and software by using cutting-edge concepts and attempt to develop these concepts further. But also strive to educate the students on our team and those in our community.

Team Management

Our team is broken into three sub-teams:

- Mechanical/Hardware
- Programming/Software
- Outreach/Business

Each member is able to be on multiple sub-teams to best suit their interests and skills. To stay organized, each sub-team has a team lead that acts as a project manager. To help the team communicate, we utilize a private Discord server for 24/7 communication!

Season Goals

At the start of this season, we had each team member set a goal for themselves, and then the whole team decided on a few goals. To achieve these goals we made them a main focus in our meetings and team interactions. Here are some of our notable goals:

Team Goals:

- ☑ Use subtractive machining (CNC Milling/Routing)
- ☑ Have every single team member learn something new
- ☑ Build a visually appealing robot
- ☑ Increase our technical outreach efforts

Achieving Goals

Throughout the season we have an overall goal of continuous growth as individuals and as a team. This pushes us to master new skills and expand our knowledge. To help achieve our team and individual goals, we leverage peer to peer mentorship and our mentors to gain skills and knowledge.

Individual Goals:

- ☑ Colin - Utilize a feedback control path follower
- ☑ Andrew - Learn about leadership
- ☑ Bren - Learn to operate and maintain a CNC Router
- ☑ Ryker - Learn something new about the engineering design process

Sustainability

Our Continuous Mentors

Leigh Elkins

Business, Outreach

Marcus Kauffman

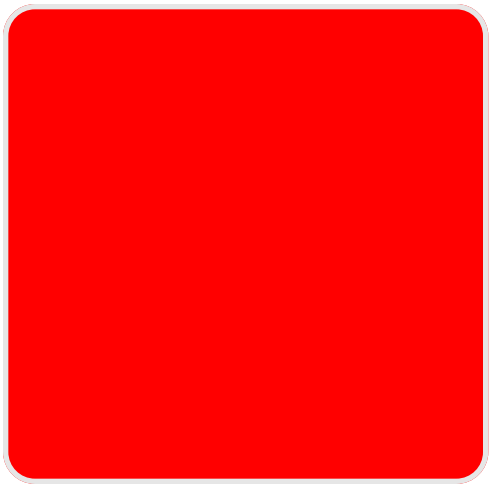
Software, Control Theory
*Student Mentor (FRC)

John Kelly - NASA

Hardware, Software

Kurt Elkins - RTX

Hardware



Learning From Mentors

Our mentors **provide guidance through complex technical challenges** by sharing their expertise in each area. Beyond just providing technical advice, our mentors and coaches help to **instill valuable life skills like teamwork, problem-solving, perseverance, and clear communication**. Through all of the knowledge passed by mentors, we are working on building our passions for STEM. This season we especially focused on learning CAD and strong design principles from our mentors.

Funding Our Team

Our community team is funded through gracious donations from sponsors. We greatly appreciate our sponsors for their new and continued support of our mission. In addition to our sponsors, we also seek grants for additional funding.

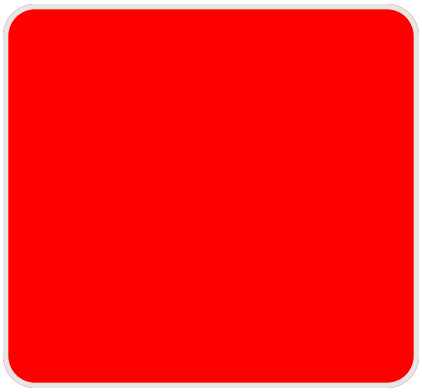
This season we are grateful for the support provided by the *Gene Haas Foundation* (1st year), *NASA* (3rd year), *RTX* (3rd year), *RTX* (3rd year), and *ASW Engineering* (2nd year).

Finding New Mentors

Finding new mentors is a challenging task. **When we are faced with a challenge that falls outside of the expertise of our continuous mentors, we seek guidance from new mentors**. These mentors are primarily students or coaches from other FTC teams through the use of the Unofficial FIRST Tech Challenge Discord server. Also through student and current mentor connections. **Each new mentor provides unique insights** to contribute to the overall success of our students.

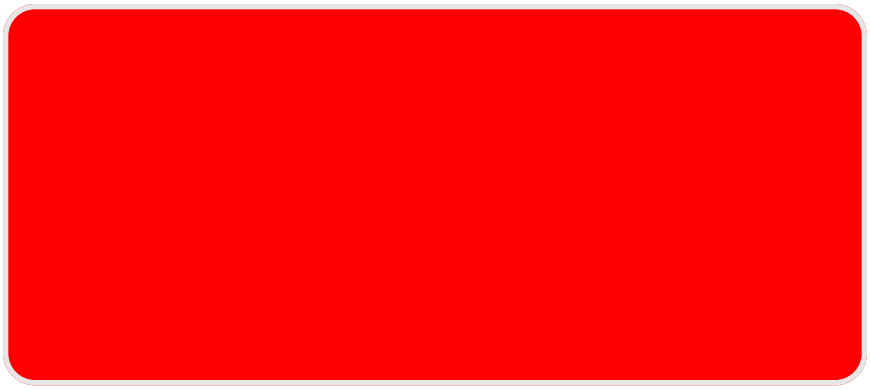
Community Outreach

≈20,000 People reached (social media and events combined)



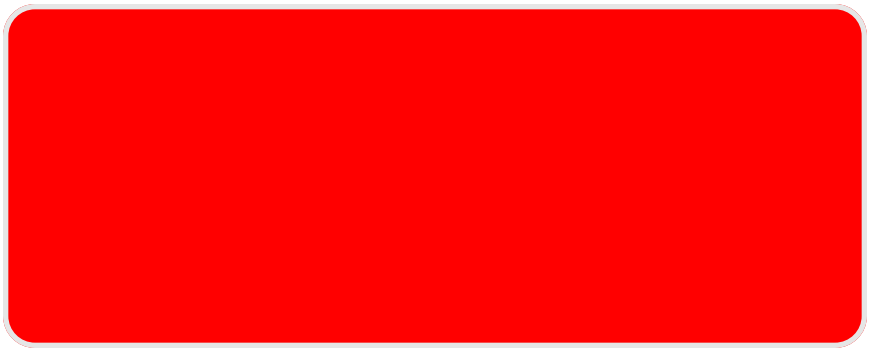
Fall Festival

Our team developed valuable skills in community engagement and working well with younger students by running the arts and crafts table at a local Fall Festival.



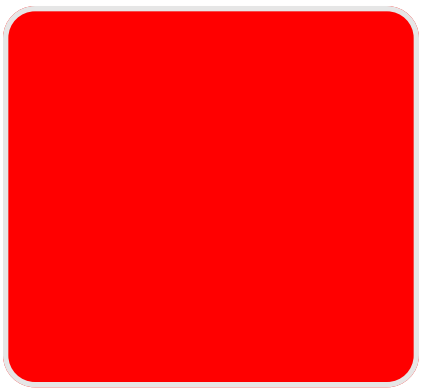
STEM Night

For the second year, we engaged with our local community by hosting a booth at the combined STEM Night for Harelson Elementary and Cross Middle School Realizing how much more memorable a give-away makes an event, we 3D printed sharks to give away to attendees. This initiative **fostered excitement about STEM field in both students and adults** alike, and provided a memorable connection to this community.



AZ School for the Deaf and Blind

After the conclusion of last season we reached out to the Arizona School for the Deaf and Blind (ASDB) and organized a time to meet and demonstrate our robot to an engineering class. The ASDB had previously planned to found their own robotics team, and **our demo provided continuing inspiration for their robotics program**. Last season we did a similar event, and learned about the importance of interactivity, so we put an emphasis on interaction!

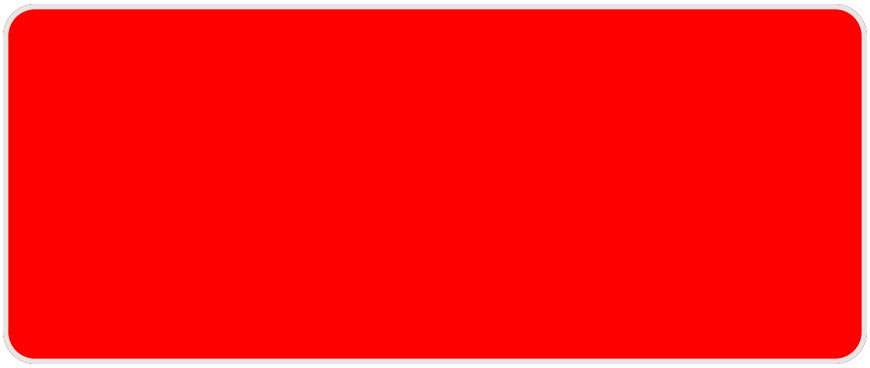


Vendor Fair

During an Arizona ESA Vendor fair, we had robot demonstrations and sold food/drinks to **help fundraise for the team while showing off robotics and how awesome STEM is.**

350+ Total man hours of team outreach

Technical Outreach



R-Team Robotics

For the past two seasons, we have had John Harkey, a representative for R-Team Robotics visit us on multiple occasions. He presented some of the amazing robots the team has built, and we introduced him to our robot and FIRST. Through his demonstrations of the team's various robots **we learned a lot about the power of iteration**, and how many iterations you need to go through for a completed product.



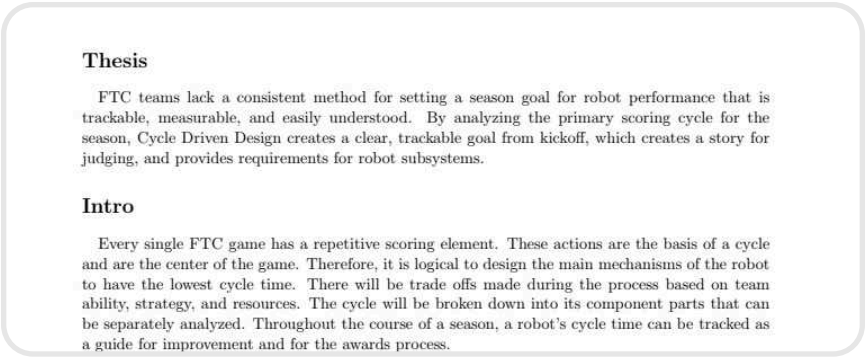
Microchip MASTERs

Over the summer we joined FIRST teams at the Microchip MASTERs conference. During the event we **spoke to engineers of various specialties to introduce them to FIRST**. We even got some design advice!



PCC B³

At the beginning of the season we attended an event at Pima Community College hosted by 15135 - Venom. We got a fantastic tour of their brand new engineering building and **learned a lot about using community college as an avenue for education**!



Thesis

FTC teams lack a consistent method for setting a season goal for robot performance that is trackable, measurable, and easily understood. By analyzing the primary scoring cycle for the season, Cycle Driven Design creates a clear, trackable goal from kickoff, which creates a story for judging, and provides requirements for robot subsystems.

Intro

Every single FTC game has a repetitive scoring element. These actions are the basis of a cycle and are the center of the game. Therefore, it is logical to design the main mechanisms of the robot to have the lowest cycle time. There will be trade offs made during the process based on team ability, strategy, and resources. The cycle will be broken down into its component parts that can be separately analyzed. Throughout the course of a season, a robot's cycle time can be tracked as a guide for improvement and for the awards process.

Samuel Kunzeman

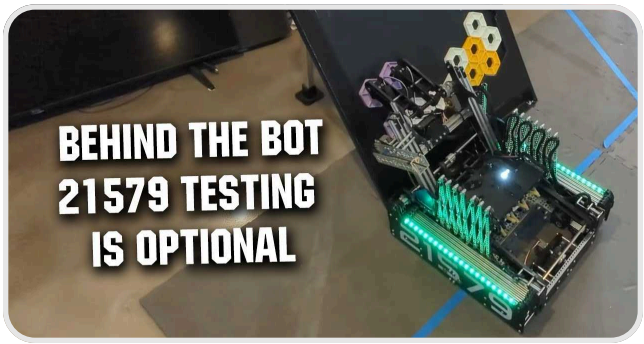
During the offseason we had the pleasure of having Samuel Kunzeman, an engineer at Caterpillar and FIRST Mentor/Volunteer speak to us about strong design principles. During this time he **introduced us to Cycle Driven Design, which is the basis of our current design process** now! The team learned a lot about design processes and about what Caterpillar does and how massive their machines truly are! Later in the season he **also provided mentoring on the portfolio**.

FIRST Outreach

≈1500% increase in people reached via social media vs last season (18k reached)

Peer Mentoring + Assisting

Starting last season and continuing into this season, we have worked with teams to peer mentor them in all aspects of FTC. We work with teams like 26037 - *Cyber Tigers* (AZ) and 5661 - *Wolves Robotics Wolfpack* (AZ) to **help them better understand mechanical, software, and portfolio design principles** to help every team be as successful and competitive as possible.

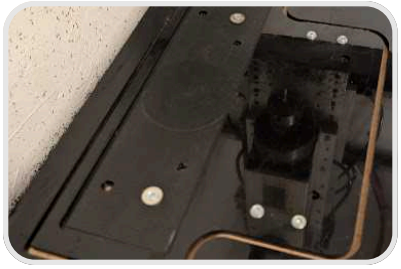


FUN Behind the Bot

Over the off-season we had the pleasure of being the **first AZ team featured on FIRST Updates Now's Behind the Bot series**, showcasing a deep dive on our robot to 1.9k viewers. This caused numerous emails asking questions, allowing us to support more teams!

Collaborative Meetings

As a goal for the season we wanted to speak with as many teams as possible and have **met on a call with 1 team/month** since kickoff, on top of **messaging with teams throughout the season (90+ teams!)**. See a list of teams in our pit!



CNC Impact

Throughout the season **we used our new CNC skills to cut parts for teams** including 14584 and 25620.

Scrimmage

To help teams get driver practice and connect, we hosted a scrimmage for 3 teams the week before the State Championship!



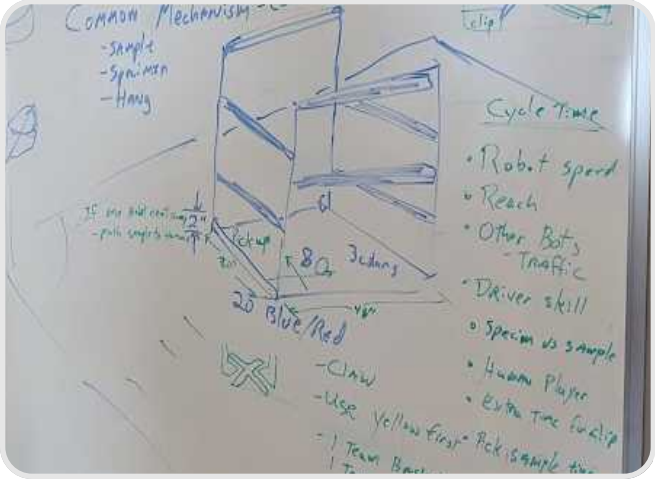
Volunteering

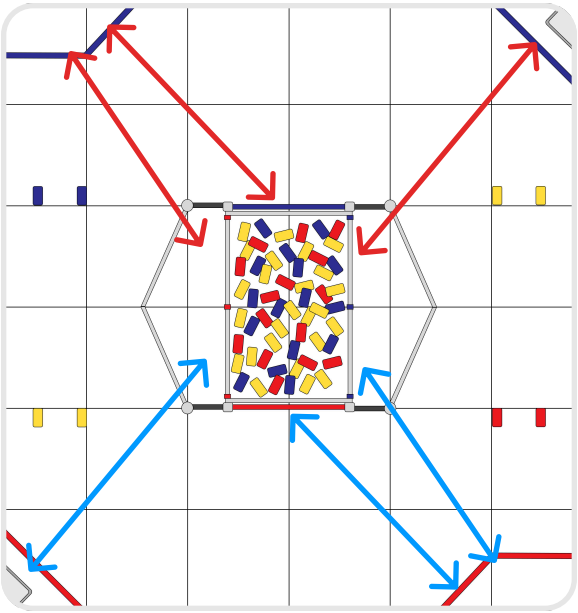
During the season we supported 1 event (2 volunteers), and had 1 member volunteer at 3 other FLL/FTC events!

Game Strategy

Samples over Specimen

Our initial season strategy focuses on optimizing cycle times for a primarily sample-focused robot and places less priority on specimen, while retaining the ability to some way to score them. This is due to the greater point weighting for samples. We also decided to prioritize the use of a **“pass-through” design, where the robot is able to score without the need to turn**, streamlining scoring cycles.





Interoperability + Three R’s

To adapt to the new playoff structure for this season, we believe that interoperability and reliability are incredibly important aspects of the game. During playoffs, both robots must play every match, so being **able to score samples and specimen reliably** is an incredibly valuable asset during alliance selection. We also have placed emphasis on the “Three R’s”: **reliability, robustness, repairability**. As we progressed throughout the season, we found the importance of these values and have continued to iterate on mechanisms like the extension with the values in mind.

Game Element Risk Analysis

Analyzing the risk associated with each game element was a fundamental way we could decide on which game element to prioritize in our design strategies. **Through SWOT** (Strengths, Weaknesses, Opportunities, Threats) **Analysis, we determined that samples had the least amount of risk**, but specimen had a greater opportunity. We ultimately chose to prioritize samples for the better opportunity cost.

Samples	Specimen
Strength • Easier to score • Easy to pick up and handle • Doesn't have to be handled by the human player • Simpler cycle	Strength • Higher point value • Doesn't interfere with sampl scoring • Utilizes alliance-specific san • Once made, quicker cycle ti
Weakness • Fewer points •	Weakness • Hard to handle • Has to be placed in a specifi position • Longer cycle time overall • Requires human player
Opportunities • 8 points	Opportunities • Higher scoring
Threat	

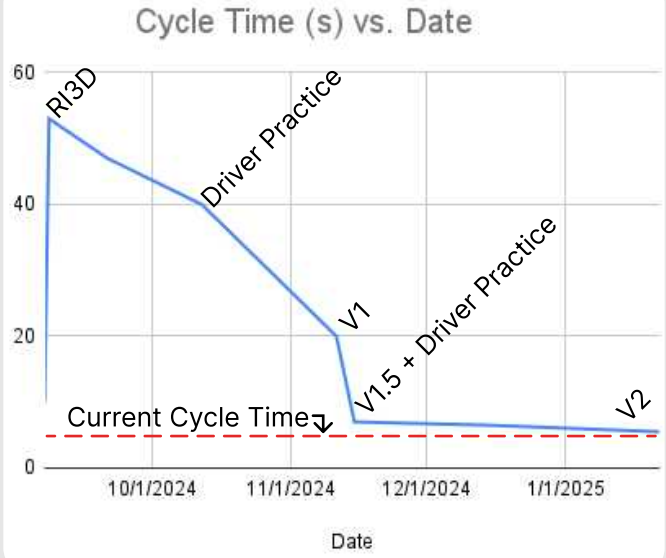
Design Process

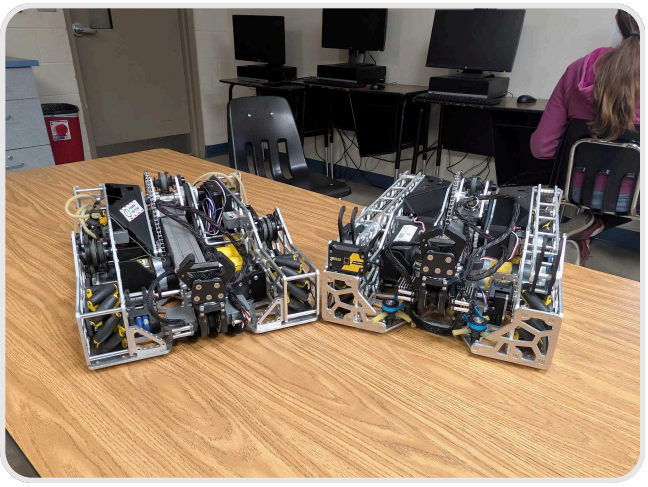
Ryker

"Working on the robot helped me understand the importance of design processes in robotics."

Cycle Driven Design

The idea of cycle driven design is to **find what systems provide a faster cycle time**. To do this we test the bot's cycle times and log them weekly to see if our design changes are making an effect on the cycle time for the bot. We also take into account driver practice, improving cycle time and how that could be the reason cycle times improve, not systems. Our RI3D robot provided a cycle time of 40s at the least, compared to our current time of 5.25s. A decrease of 86.25%.





Dual Bot System

This season we decided to run two bots in parallel. One bot is considered the "active" bot and is the bot our drivers are practicing on and the software team is working on for autonomous and driver enhancements. The other bot is considered "inactive" and is used to test new system by the build team. This practice helps us **reduce the risk of not having a functional robot** at competitions.

CAD

Imagine the system and draw it in complete detail using Autodesk Fusion

Prototype

3D Print parts and do a test assembly to see if the idea works and is possible to build

Revise

Evaluate the prototype for failure points, weaknesses and improvements

Implement

Add the component to the non-active bot to integrate it with other systems

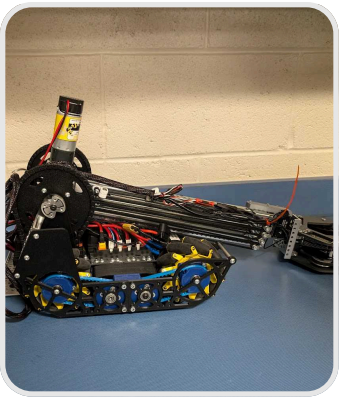
Swap

Swap out the bots and equalize the now non-active bot to add more features

Grace

"Learning SketchUp from Mr. Kelly introduced me to 3D modeling applications, and encouraged me to learn more about CAD."

Meet MAKO



V0: The very first iteration of the robot that we used as a proof of concept by **assembling and programming the bot in 3 days**. ~40s cycles (with practice).

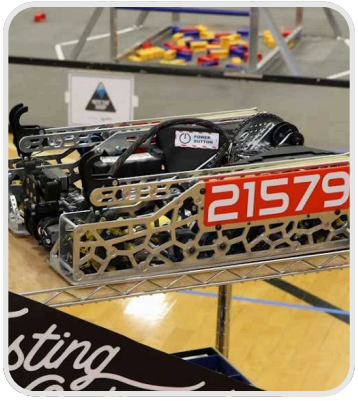
Pros: Simple, quick proof of concept, easy to test on

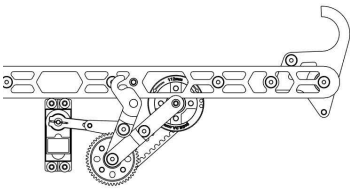
Cons: Hard to control in code (pgs 10, 13), slow cycle time, hard to drive, wrist mount breaks often (pg 11), no localization (pg 9), easy to tip

V1: Using issues from the first iteration, we used CAD to develop a fully custom robot that is more developed than V0. ~20s cycles (no practice yet).

Pros: Simple to control in code (pgs 13), increased wrist capabilities (pg 11), consistent extension distances (pg 10)

Cons: Extension stringing fails often (pg 10), lots of pivot backlash causing inaccurate angles (pg 10), no ascent





V1.5: Added our ascent mechanism and made **significant software changes to improve in both TeleOp and Autonomous** (pgs 14, 15). Lots of reliability and repairability issues with the extension and ascent mechanisms. ~6.5s cycles.

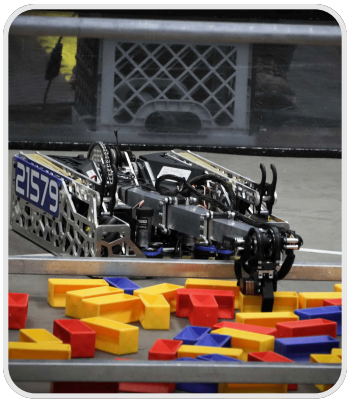
Pros: Can hang (pg 12), driver automations implemented (pg 14)

Cons: Hang arms not deploying (pg 13), extension strings snap often (pg 10), wrist structure breaking/stripping (pg 11)

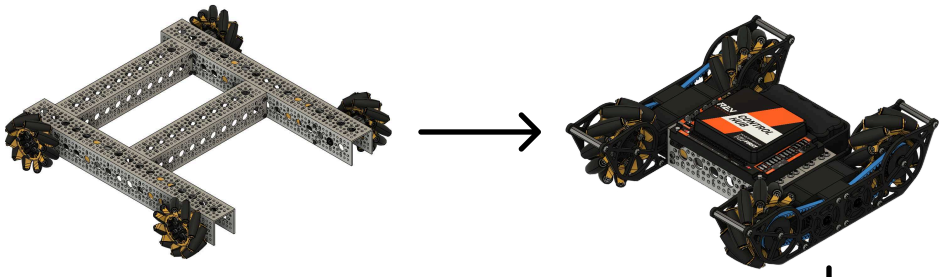
V2: Tried an active intake for specimen, that we decided was not optimal and removed. Most changes were **reliability, repairability, and robustness modifications**, as well as improving our extension speed and strength. ~5.5s cycles.

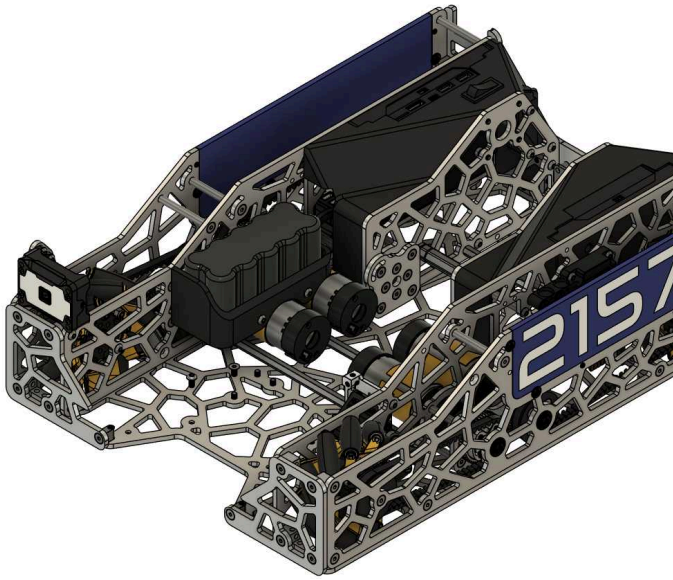
Pros: Faster extension (pg 10), easy to repair extension (pg 10), reduced pivot backlash (pg 10), motion profiling (pg 13)

Cons: Can grab opposing alliance game elements



Drivetrain





Overview

Custom Parallel Plate Design - Prioritizes simplicity and is compact, allowing increased space for more mechanisms, like the power take off (pg 12). Designed and machined in-house.
4mm 7075 Aluminum Plates - 7075 alloy chosen for its near double strength (83k psi) compared to the more common 6061 alloy (42k psi).
Hand Pocketed - Provides a **~70% reduction** in weight and increases aesthetic appeal
1:1 Gear Reduction - Allows for simplicity and easy RPM management within software
Unique Shape - With the sizing of the extension and pivot mechanisms in mind (pg 10), we opted for a longer rectangular shape to best fit them.

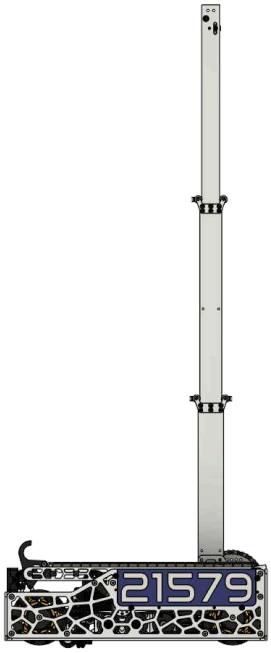
Repairability

Having the ability to easily make repairs to the most important aspect of the robot, being the drivetrain, is incredibly important. As we designed the robot in Autodesk Fusion, we made as much of a great effort to make the **replacement of motors, belts, and wheels require only one size wrench** through the use of the same M4 sized screws across the entire drivetrain.

Localization

goBILDA Oriented - We utilize the goBILDA localization ecosystem to stay oriented, using two of the 4-bar odometry pods and a pinpoint computer for the upgraded IMU.
Pinpoint Position - Utilizing the goBILDA Pinpoint allows us to read our odometry pods at 1500hz compared to the control hub's 30-90hz, **improving consistency and acceleration localization**
Tried and True - To determine whether to stick with custom or use this COTS setup for odometry, we put them head to head and found the COTS approach to have closer to ~0.05in of accuracy compared to the ~0.1in of accuracy of the custom solution.



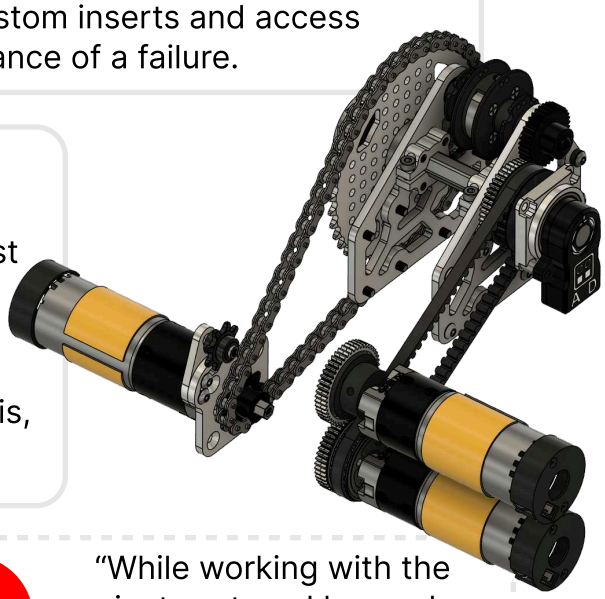


Custom Boxtube Extension

Unique & Compact Design: To save space and stay within size constraints, we use 1.25”, 1.5”, and 1.75” aluminum box tubes cut in house to extend to 45” max. Allows for significant space savings compared to standard slides.
Smooth and Stable: Using 24 bearings both on the inside and outside, we ensure smooth and stable motion from the extension.
Quick and Powerful: Driven by two 1150rpm motors geared up 1.5:1 powering a 16mm spool, allowing for quick movement, full extension in 0.6s.
Robust Stringing: Uses cascading rigging that allows each stage to move in unison, eliminating the need for end stops. Custom inserts and access holes allow for repairability and reduce the chance of a failure.

Pivot Mechanism

Strong and Fast: As a design requirement, we needed a fast pivoting mechanism, so we utilize a 223rpm on a 4.2:1 reduction (~53rpm effectively)
Always in the Know: Utilizing an East Loop Components absolute analog encoder, we always know where the pivot is, even after a power cycle.



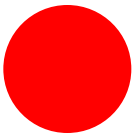
Iteration & Improvements

Extension:

1. Viper Slides to Telescoping Box Tube
   a. Was too heavy + hard to control
2. Increased Extension Speed (2 motors)
   a. Wanted to decrease extension time
3. Improved Stringing Inserts
   a. Old inserts allowed the string to slip out easily
4. Added String Tensioner + Access Holes
   a. Allows easy tensioning and access for repairability, reducing failures

Pivot:

1. Implemented Chain Tensioner
   a. High backlash was causing inaccurate control



Noah

“While working with the pivot system, I learned the importance of how to fit systems in constrained spaces.”

Variable Feedforwards

Gravity is an important factor in controlling mechanisms. To account for gravity and friction while using our motion profiles (pg. 13) we vary our feedforward term of the PIDF loop using the math below.

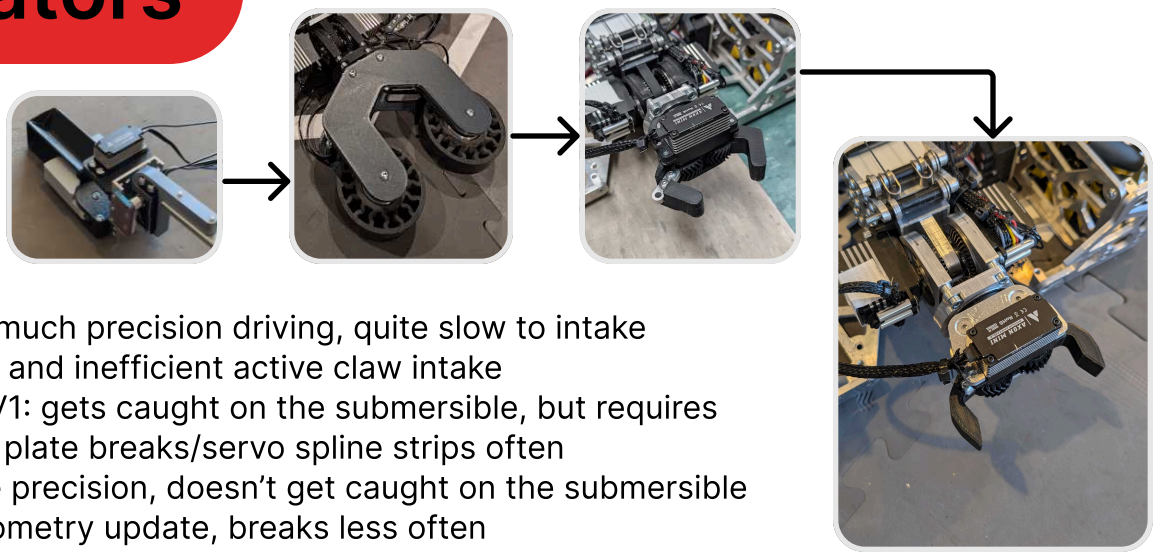
Extension Kf:

$$(K_{gs} + K_{gd} \cdot x_e) \cdot \sin(\theta_p) + K_s \cdot \text{sgn}(v_e) + K_v \cdot v_e + K_a \cdot a_e$$

Pivot Kf:

$$(K_{gs} + K_{gd} \cdot x_e) \cdot \cos(\theta_p) + K_v \cdot v_p + K_a \cdot a_p$$

Manipulators



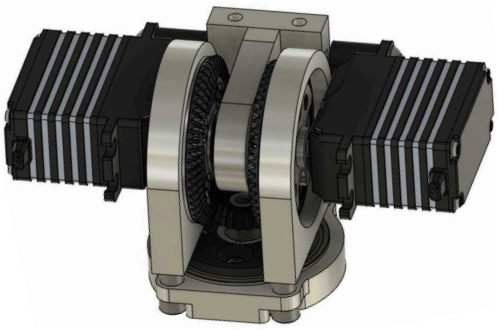
Claw

V1 - Required too much precision driving, quite slow to intake
V2 - Heavy, bulky, and inefficient active claw intake
V3 - Redesigned V1: gets caught on the submersible, but requires less precision, top plate breaks/servo spline strips often
V4 - Requires little precision, doesn't get caught on the submersible easily due to a geometry update, breaks less often

Floor Specimen Intake

As an attempt to improve specimen cycle times, we implemented an active intake like the one you see to the right. **After comparing the efficacy of the active intake vs intaking specimen off the wall, we decided to remove the active intake** in favor of intaking off the wall. The intake time with this mechanism was about ~6s compared to the ~3s of the claw and wall method.





Differential Wrist

2 Degrees of Freedom: Using two degrees of freedom allows us to grab samples in any orientation (pitch, yaw).
Aluminum Frame: After learning that 3D prints did not suffice (failing 4 times throughout the season), we migrated to a custom machined aluminum frame.
Quick and Powerful: To increase speed, we replaced the previously used goBILDA Torque servos with Axon Max+ servos. This change allows the wrist to reach any position in just 0.75s, while maintaining the force need to hold positions.

Kinematics

To help us calculate and derive the positioning of the wrist, we utilize kinematics. We use the equations to the right, where θa is the pitch of the wrist, θt is the yaw of the wrist, and m1/m2 represent the positions of the servos. You might notice that these equations only calculate the current position, so we can use algebra to rearrange the equation in code to figure out the necessary servo positions for the desired output.

$$\theta_t = \frac{m_1 + m_2}{2}$$

$$\theta_a = \frac{m_1 - m_2}{2}$$

Bren

"Designing the PTO taught me about the effectiveness of a 4 bar linkage for stability"

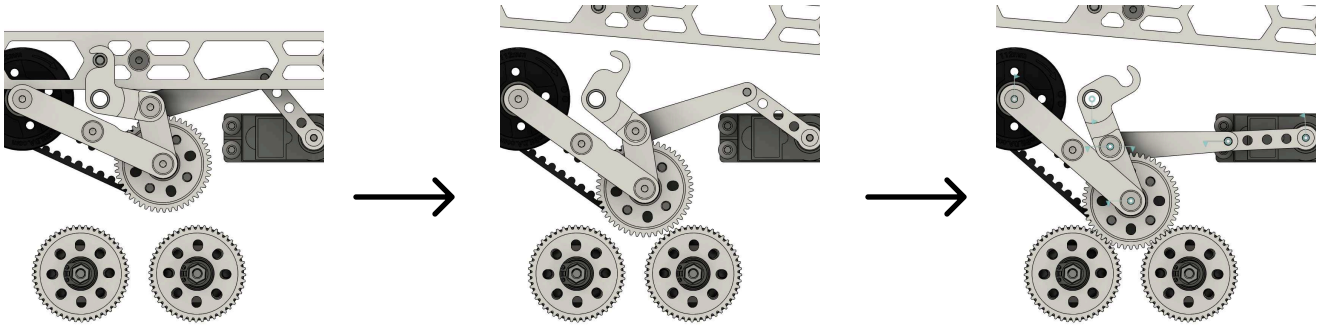
Ascent/Hang

Drivetrain Power Take Off

Our innovative Power Take Off (PTO) allows us to transfer the power of the drivetrain to our hang spools. We use one PTO on each side of the robot, each driven by a goBILDA torque servo. The two sprung loaded hang arms are released when the PTO engages. Driving the drivetrain motors causes the spools to start winching down on the hang arms.

Why a PTO?

One of our goals was to have a simple and elegant robot with innovative features. We wanted to **avoid adding additional actuators** to our robot which would add more mass and take up more space. Utilizing a PTO allows a new subsystem to be controlled using primarily actuators from a separate subsystem, achieving this goal **in a robust and resilient manner.**



The PTO is disengaged. During this state the hang arm is locked in position and the robot will drive as normal without worrying about the hang arms flipping up.

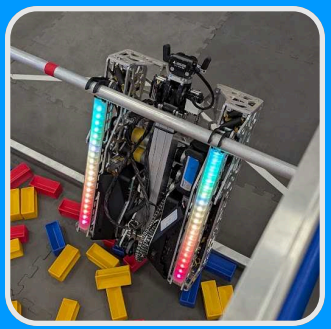
The PTO is half engaged. This step of the process allows the hang arms to fling up and into position. Since the gears aren't engaged, driving will continue as normal and the hang arms will not retract.

The PTO is fully engaged and the linkage is over-centered to prevent the PTO from disengaging. The spool for the hang arms is now powered by the drivetrain motors and hanging can commence.

Analyzing Ascent Risk

Climbing with a 15kg robot is a challenging task, so assessing the risks involved is important. By doing **cost-benefit analysis on the various ascent levels** (1-3), we determined that we would **not continue past level 2**. This determination was made based on a few points:

- Level 2 Ascent: ~2 seconds (~7.5 pts/s)
- Level 3 Ascent: estimated ~15 seconds (~2 pts/s)
- Additional cycles: ~1.6-2 pts/s + reduced risk of robot damage



Control Theory

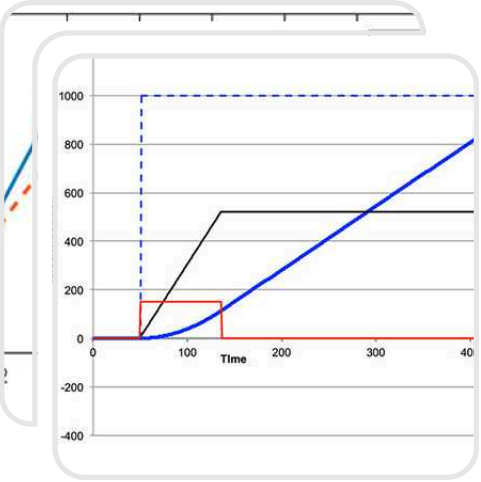
Colin

"Control theory proved to me that math and physics have useful applications in robotics"

PIDF Controller

To aid in control of both our pivot and extension subsystems, we utilize a PIDF controller for each. PIDF works to "close" a software loop, working solely with the information provided by sensors. The F term (feedforward) "re-opens" the loop by predicting the forces needed to overcome gravity and friction. We utilize this term to work against gravity, since gravity is a constant. When combined with the rest of the PID controller, we essentially have a **closed loop system while having compensation for gravity**.

$$u(t) = K_p(e(t) + \frac{1}{K_i}\int_t^0 e(\tau)d\tau + K_d\frac{d}{dt}e(t)) + K_f$$



Motion Profiling

To aid in keeping our motion smooth and consistent, we utilize asymmetrical motion profiles. These use a combination of PIDF and a function of time to **limit the velocity and acceleration of a system. This allows for fast but consistent motion of the pivot and extension systems**. We estimate that this will reduce wear by ~70% by modifying the rates of acceleration on both side of a motion. The math and theory behind the profiles can be found in our pit.

Commands + State Machines

To aid in writing effective and easily readable code, we utilize the FTCLib command base paradigm within our code. What this allows us to do is create "commands" that act as state machines. These commands can be a single mechanical event or a sequence of events (e.g., autonomous). Using this paradigm allows us to **create sequences of events significantly easier and makes iteration on autonomous faster**. We utilize four primary state machines that affect driver automation (pg 14) and autonomous:

- Game Element (Sample, Specimen)
- Scoring Target (high basket, low basket, specimen shuttle, specimen deposit)
- Bot State (intake, deposit, ascent)
- Alliance (blue, red)

Johnny

"While working on wiring the feedback LEDs, I learned about the importance of insulating wires."

Driver Control

State Machine Automations

Intake: Using a single button press, the driver can initiate our intake sequence. This sequence varies, but in essence: attempt to grab a game element, *detect whether the element is now held using analog encoder in the claw*, if not then reset, if so then set the bot for depositing.

Deposit: With the same button as intake, the driver can deposit a game element depending on the current state. The sequence attempts to score the element (dropped in the basket or scored on a chamber) and then resets the robot to be ready to intake more elements.

Ascent: Also a single button press, a sequence can be initiated by the driver to begin our ascent. This is done by resetting all the subsystems to the appropriate position, releasing the hang arms, engaging the PTO, ascending, and then locking the PTO once ascended.

Autonomous in TeleOp

As a part of our deposit automation, the driver can enable automatic driving to score specimen. **This allows the driver to put their controller down while the robot intakes specimen from the observation zone and scores them!**

Safeties

Driver safeties allow our drivers to keep the robot in one piece and help prevent them from incurring fouls.

Extension Limits: To alleviate problems stemming from encoder drift and mechanical backlash, we automatically apply a small negative power to the extension motors and reset the motor encoder when the extension returns to the zero position. This makes the encoder reading always accurate to prevent exceeding the 42" extension limit.
Protection Limits: To protect the robot, each subsystem has software limits using motor encoders and servo position clamping that prevents the driver from exceeding the mechanical capabilities of the system.

Feedback

Ensuring the driver knows what is happening with the robot is important. To do this we utilize the gamepad's built-in LEDs and the rumble/vibrations on the controller, as well as LEDs on the robot.



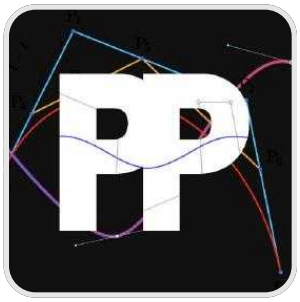
Why Does It Matter?

Each of these elements make up a cohesive control scheme for our driver that **reduces the chance of error and helps optimize cycle times**. A large part of optimizing our cycle times comes with driver practice, and if the sequences can be automated and optimized, the driver requires less practice for the same task. These automations combined with driver practice has yielded about ~27 seconds of cycle time reduction.

Autonomous

Pedro Pathing

To follow our paths, we utilize **Pedro Pathing**. This algorithm utilizes 3 individual PID controllers (heading, translational, drive) and Bezier curves to follow paths accurately. The **PID controllers allow the robot to adapt if it contacts another robot** or gets stuck somewhere. We found that Pedro Pathing is significantly more effective in consistency (50% more consistent) than Roadrunner through empirical testing processes.



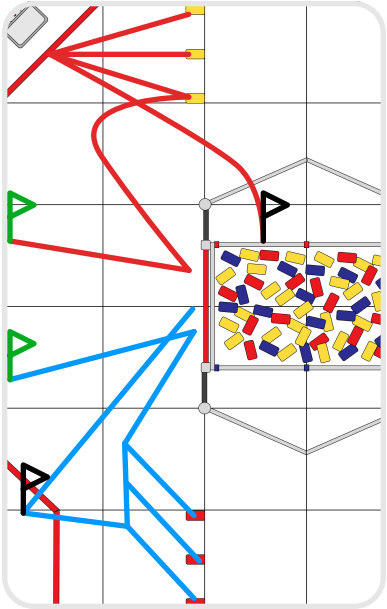
Our Autonomous

Sample Auto (35 or 37 pts):

1. Score preloaded specimen on the High Chamber OR preloaded sample in the High Basket
2. Intake + Score the Placed Neutrals in the High Basket (x3)
3. Achieve Level 1 Ascent

Specimen Auto (53 pts):

1. Score preloaded specimen on the High Chamber
2. Move all 3 alliance specific samples into the Observation Zone and form specimen
3. Intake + score each specimen on the High Chamber (x4)
4. Park in the Observation Zone



Field Configurable

As we place the robot on the field to begin a match, we have the ability to slightly modify our autonomous paths using the gamepad connected to the driver station. For example, we are able to select whether we are preloading a sample or a specimen into the robot, **effectively modifying the path to work best with our alliance partner**. We also use this time to specify which alliance we are on to be used later for driver safeties.